PYTHON AND AI · EPISODE 01

Why Learn Python Now?

Course Introduction

KEY POINTS

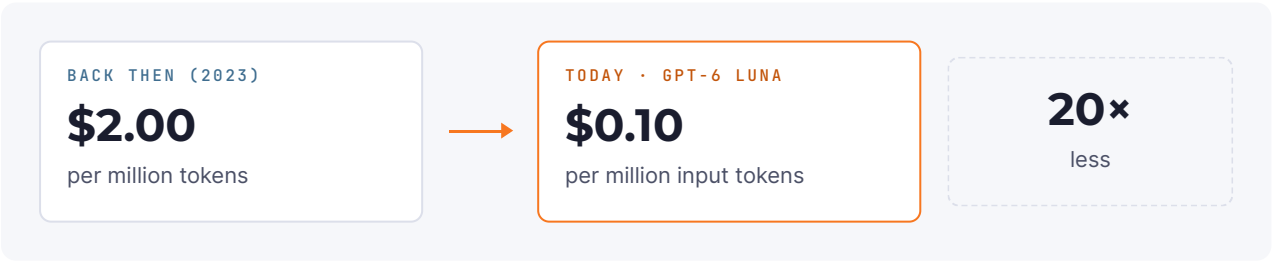
- You don't need to train a model anymore. **You can just call one**, and prices keep falling.
- Apps use AI in three ways: **chat apps**, **cloud APIs**, and **local models** on your own computer.
- Python is **#1** on the TIOBE index (November 2026, 20.86%) and powers nearly half of all new AI repositories on GitHub.
- Six stages: two to learn the basics **by hand**, then four projects where AI coding tools join in.

01 The Moment Everything Changed

▶ VIDEO 00:00

DATE	WHAT HAPPENED
Nov 30, 2022	OpenAI releases **ChatGPT**, built on the GPT-3.5 model.
Mar 1, 2023	ChatGPT's model opens to developers through an **API**, ten times cheaper than OpenAI's earlier GPT-3.5 models.
Jan 20, 2025	**DeepSeek-R1**: on par with OpenAI's o1, open source (MIT License). A week later its app is #1 on Apple's App Store in the US.
Today	Open models you can run yourself: Meta's **Llama**, Google's **Gemma**, OpenAI's **gpt-oss**, **DeepSeek-R1**.

FIGURE 1.1 **Prices kept falling**



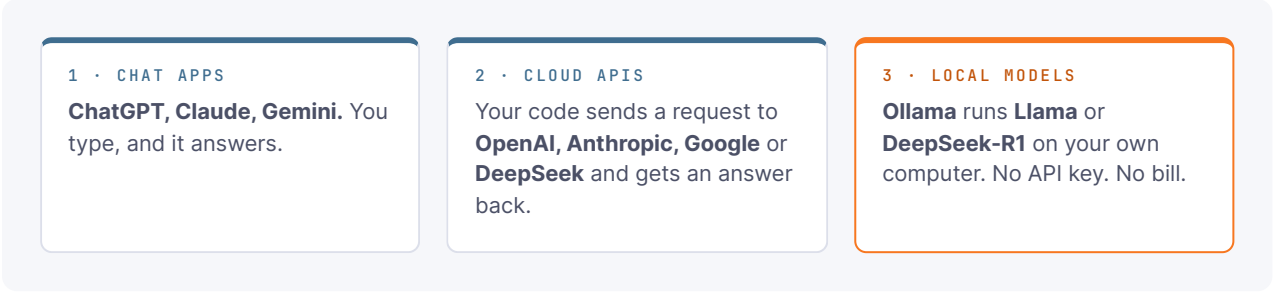
A token is a small piece of text, often part of a word. Cheap APIs and free open models opened AI development to every developer.

02 How Apps Use AI

▶ VIDEO 02:03

Adding AI doesn't mean building your own model. The software a company already runs **starts calling a model**: the app stays the same app, it just gets a brain. The door that's wide open now is **building applications that use AI**.

FIGURE 1.2 **Three ways to use a model**



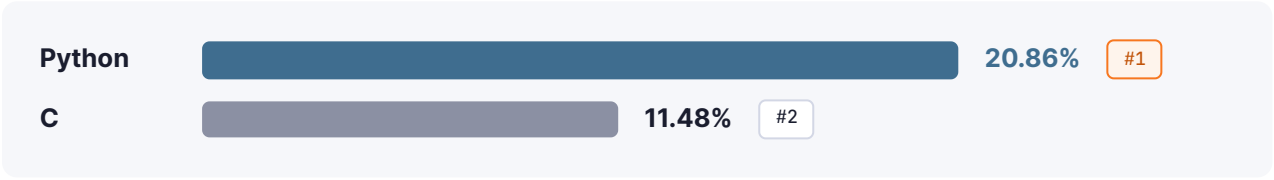
In this course you'll learn how these work, and how to call them from Python.

03 Why Python

▶ VIDEO 03:16

According to GitHub (Octoverse 2025), Python powers **nearly half of all new AI repositories**. The TIOBE index ranks languages by popularity every month:

FIGURE 1.3 **TIOBE index, November 2026**



Nearly double. In 1996 Python averaged twenty-seventh place; today it's first, and AI is a big part of that story.

04 What You'll Build

▶ VIDEO 03:56

FIGURE 1.4 **Six stages: the foundation, then four projects**



Every technique shows up inside a real project, so you know why it exists, not just how to type it. Each project stage ends with where to go next.

05 Why It's Built This Way

▶ VIDEO 05:14

- **AI coding tools join in the project stages:** they help you write code, add comments, explain tricky parts, and make programs better.
- **Professional habits from day one:** handling errors, logging, and debugging. Writing code is one skill; finding and fixing problems is another.

06 Who It's For · How to Learn

▶ VIDEO 06:10

From zero: for complete beginners, anyone curious about AI, scraping or data, career switchers, and students preparing for exams, competitions or interviews.

> **In the first two stages, don't use AI coding tools. AI can only amplify what you already understand.**

Type every example yourself and make the mistakes yourself; that's how the basics stick. The mistake from the video, and what Python says:

hello-typo.py — PYTHON

```python
# This file raises an error on purpose: a typo ("nme" instead of "name") causes a NameError.
# Fix the typo and run it again.
name = "Ada"
print("Hi, " + nme)
```

Output

```
Traceback (most recent call last):
  File "hello-typo.py", line 4, in <module>
    print("Hi, " + nme)
                  ^^^
NameError: name 'nme' is not defined. Did you mean: 'name'?
```

TRY IT **Type it. Run it. Break it. Fix it.**

Change `nme` back to `name` and it prints `Hi, Ada` (`hello.py` in the code folder). Watching feels like learning. Typing is learning.

CODE 3 files on GitHub github.com/abitclearer/python-and-ai/tree/main/01-why-learn-python-now